

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

The only place in the civilised world without rats, galactic fireworks, and quantum materials.



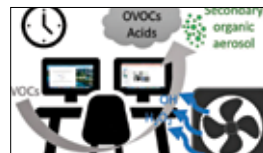
Good to go

Want to predict the most mind blowing space headlines? Track the astronomers that have been allocated scope time on the JWST, and find out what they are going to study about the mentioned exoplanets, black holes, or galaxies, then find out what makes these targets unique. <https://dgit.in/jwstgo>

WHAT'S NEW

Deodorization tech can release unintended pollutants

There are two types of air purifiers in the market, one that use mechanical filtration by passing the air through membranes, and electronic cleaners that include ionisers, plasma sterilizers, and photocatalytic oxidation purifiers. These release compounds that react to other compounds in the environment, breaking down pollutants. This study focused on a type of electronic air purification devices known as hydroxyl generators, which are used to break down odour in the environment. The equipment is commonly used to remove odour buildup by pets, mould and



radicals released by the generator reacted with indoor organic compounds and quickly formed organic amino acids, as well as organic aerosols, that could have harmful effects. The researchers warned that customers may not be aware of the additional chemistry that takes place in the environment. <https://dgit.in/secoh>

smoke. The devices are also used in office complexes and restaurants. The research shows that the hydroxyl

Starlink constellation used for geolocation

Researchers have used the signals from the Starlink constellation of satellites and systems similar to the GPS network. The technology works by carefully timing signals simultaneously to multiple satellites in the constellation, know their exact positions and movements, and then figuring out the accurate position of the receiving antenna on the ground. Note that the researchers had no direct



access to the actual contents of the communications, only data that could be used to infer the satellite's location and movement, and did not need assistance from SpaceX for the experiment. The geolocation

provided by this jerry rigged system was accurate to within five meters. For comparison, the resolution of the GPS system is around 3 meters, the Galileo constellation offers a sub meter resolution, and the BeiDou constellation offers a resolution of 3.6 meters. The more satellites SpaceX launches into the constellation, the more accurate the system can become. <https://dgit.in/slps>



Glashow resonance spotted

A highly energetic particle produced from extreme physics has been detected by the Ice Cube experiment in Antarctica, confirming a 60 year old prediction by Nobel laureate Sheldon Glashow, further confirmation of the Standard Model. <https://dgit.in/glare>



Unuclear physics

Neutrons are named for not binding with anything else, including each other, but research shows that under specific conditions, they can communicate with each other forming an unparticle. An energy carrying fuzziness may cause some neutrons to form an unucleus. <https://dgit.in/unp>



Super corals

A hydrology professor looking to track down sources of nutrient pollutants in the seas was able to successfully pin down land based sources, but also serendipitously came across "Super Corals", that can live and thrive in extremely toxic waters. <https://dgit.in/supercorals>